7	**(a)**	density in solids and liquids similar	M1	
		spacing in solids and liquids about the same	A1	
		density in gases <u>much</u> less as spacing in gases much greater	B1	[3]
	(b)	density = mass / volume	C1	
		mass = 1.67×10^{-27} kg and volume = $4/3\ \pi\ r^3$	C1	
		density = $(1.67 \times 10^{-27}) / 4/3 \times \pi \times (1.0 \times 10^{-15})^3$		
		= 3.99×10^{17} kg m^{-3}	A1	[3]
	(c)	atoms / molecules composed of large amount of empty space / nucleus has very small volume compared to volume of atom / space between atoms in a gas is very large	B1	[1]